

module 1:

Q.1 State and define four fundamental characteristics of telecommunication / data communication system.

→ 1) Delivery:
 The system must deliver the data to the correct destination device. The data should be received by the intended receiver and not any other device.

2) Accuracy:
 The data delivered should be without any errors. The data / information received should be same as the information sent, without any alterations, loss or corruption during data transmission.

3) Timeline:
 Data should be delivered in acceptable time frame. Late delivery is useless for real world application such as video conferencing, online gaming or live streaming, etc.

4) Jitters:
 Jitter refers to the variation in packets arrival time. If the packets arrive at uneven distribution / time intervals it can cause distortion in audio & video communication.

Hence it should be minimal.

Q.2

Discuss importance and application of telecommunication:

→ Telecommunication plays an essential role in many areas of everyday life such as:

- (1) Booking of hotels by travel agencies
- (2) Aviation, booking of tickets
- (3) Banking, ATM, telebanking
- (4) Government operation such as taxation
- (5) Sales, wholesales and order-handling
- (6) Credit card payments.

Importance:

(1) Global Connectivity:

connects people, organizations and countries worldwide enabling instant communication regardless of distance.

(2) Economic Growth:

Business uses telecommunication for online transaction, e-commerce, banking, marketing and co-ordination between the branches

(3) Supports modern technologies:

Technologies like internet, mobile network, etc rely on telecommunication system.

(4) Social development:

Social awareness, digital education and access to services even in remote areas.

⑤ ~~Real time~~

⑤ Faster information exchange:
 Real time communication helps in decision making in industries, education, healthcare and government operations.

Q. 3 Discuss different forms of data representation

(Text, number, image, audio, video).

→ ① Text:

Text is data represented using character encoding systems like ASCII. Each character such as letter, number or symbol is converted into binary form for processing and storage. Used in messages, emails, etc.

② Number:

It is represented in binary form (0's and 1's) inside a computer.

Can be stored as integer or floating point numbers.

Used in calculations, banking systems, scientific computations and statistics.

③ Image:

Image is a grid of pixels (picture element). Each pixel has assigned binary value according to the color and intensity.

Image quality depends on resolution and color depth

(4) Audio:

Audio is ~~is~~ generated by converting sound waves into digital signal by sampling. This process includes sampling, quantization and encoding.

Used in voice calls, music files & recordings.

(5) Video:

Video is a sequence of individual images displayed rapidly.

Each image is digitally encoded along with synchronised audio data.

Used in video calls, movies, etc.

Q.4 State key features of different generations (1G-5G) of telecommunication technology.

→ Year	Generation	Key feature
1980's	1G	- Voice signal - Analog cellular phones - NMT, AMPS
1990's	2G	- Text messages - Voice signal - Data signal - CDMA, TDMA, EDGE

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- video signal

- Voice signal

- Data signal

- W-CDMA.

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Higher data rate

Interoperability protocol

mobile IP.

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Increased
connectivity

lower latency

greater trans-

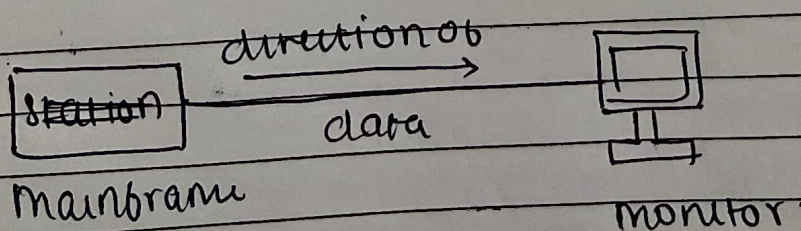
mission speed.

Q.5 Explain different modes of data flow with an example each.

Simpler:

implex:
The communication is unidirectional.

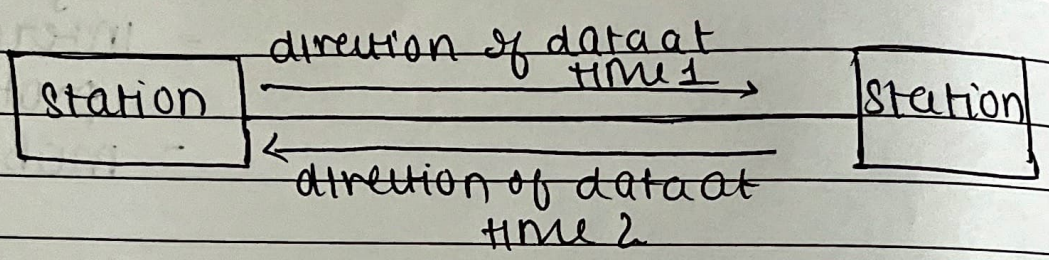
eg: keyboards (I/P) & additional
of monitors (O/P).



b) Half duplex:

Each station can both transmit & receive but not at the same time

for eg: walkie talkies, citizen band radios.



c) Full duplex:

Both stations can transmit and receive simultaneously

eg: the telephone network.

